

On the road

The building of roads goes back thousands of years. Ancient Sumeria, Egypt, the Roman Empire, and the Incas all made paved roads. Ancient Islamic cities even made roads using tar. However, it was not until the Industrial Revolution (see pp.52–53) and the motor car (see pp.92–93) that the world saw a major explosion in road building. Today, there are more than 40 million miles (65 million km) of road in the world.

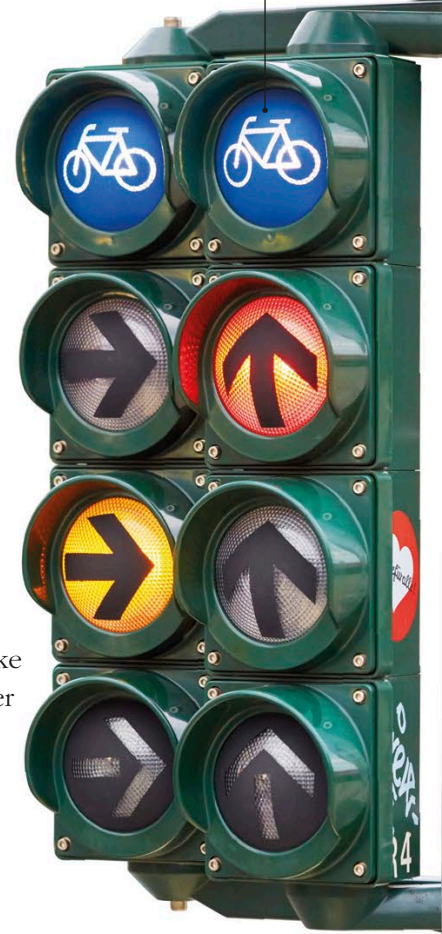
FAST FACTS

- Most roads are laid with tarmac, which is a mix of bitumen and tiny stones called aggregate.
- Roads are now often made with concrete, which costs more than tarmac to lay but lasts longer.
- Every year, 112 million tons of bitumen is used to make roads and airport runways.

TRAFFIC LIGHTS

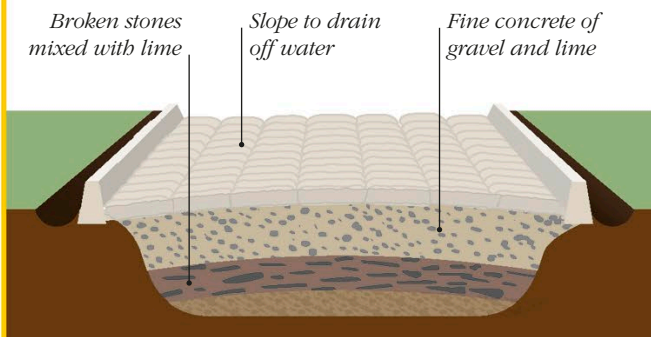
Traffic lights are vital for controlling the movement of traffic and ensuring safety at busy junctions. In the early days of automobiles, drivers relied on hinged signals and traffic police. Then, in 1912, in Salt Lake City, Utah, the American electrician Lester Wire installed the first colored electric traffic lights. Now, typically, traffic lights use LEDs (see p.181) controlled by computer systems.

Lights are programmed to turn on and off to control traffic flow at every junction.



EARLY ROADS

The ancient Romans were great pioneers of road building. To ensure the swift movement of their armies and trade around their empire, the Romans replaced winding tracks with 50,000 miles (80,000 km) of long, straight road paved with stone. They would have a slope in the middle so rainwater ran off properly.



Cross-section of a Roman road, a forerunner of modern roads

CAT'S EYES

As the story goes, while the British inventor Percy Shaw was driving home one night in 1933, he saw the eyes of a cat glowing in his headlights. It gave him the idea of reflectors—now called cat's eyes or studs—to show the center of a road at night. Modern studs often have solar-powered LEDs which glow all night.



Cat's eyes on a road in the UK